

 Did You Know?

Search algorithms help AI systems find the best solution or shortest path from one state to another.



What are Search Algorithms?

Search Algorithms are techniques used in Artificial Intelligence to find a solution to a problem by exploring different possible paths. They are widely used in games, robotics, navigation systems and intelligent software.



Why do we use Search Algorithms?

- Find shortest path
- Solve complex problems
- Support decision making
- Optimize solutions
- Improve AI performance



Types of Search Algorithms

Algorithm	Description
BFS	Explores nodes level by level.
DFS	Explores one branch deeply before backtracking.
Uniform Cost Search	Selects the path with the lowest cost.

Greedy Search	Chooses the path that seems best at the moment.
A* Search	Uses path cost and heuristic to find the best solution.



Breadth First Search (BFS)

BFS visits all neighboring nodes first before moving to the next level. It guarantees the shortest path in an unweighted graph.



Depth First Search (DFS)

DFS explores one path completely before returning and exploring another path.



Uniform Cost Search (UCS)

Uniform Cost Search always expands the node with the smallest path cost. It guarantees the optimal solution.



Greedy Search

Greedy Search selects the node that appears closest to the goal using a heuristic function. It is fast but does not always find the optimal solution.



A* Search

A* Search combines path cost and heuristic value to find the shortest and most efficient path.



Example

Start

↓

A

↓

B

↓



Comparison of Algorithms

Algorithm	Optimal	Complete
BFS	Yes	Yes
DFS	No	No
Uniform Cost Search	Yes	Yes
Greedy Search	No	No
A*	Yes	Yes



Applications

- Google Maps Navigation

- Game Playing
- Robotics
- Path Finding
- Network Routing

Common Mistakes

- Confusing BFS with DFS
- Thinking Greedy Search is always optimal
- Ignoring path cost in UCS
- Using the wrong heuristic in A*



Interview Questions

Q1: What is BFS?

Breadth First Search explores nodes level by level.

Q2: Which algorithm explores depth first?

DFS.

Q3: Which search algorithm guarantees the lowest path cost?

Uniform Cost Search.

Q4: Which algorithm uses both cost and heuristic?

A* Search.



Quick Revision

BFS Level by Level